

Qualification Pack



Lead Business Analyst

Electives: Logistics and Supply Chain Analytics / Marketing & Retail Analytics

QP Code: MEP/Q1503

Version: 1.0

NSQF Level: 5

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MEP/Q1503: Lead Business Analyst

Brief Job Description

Lead Business Analyst qualification equips learners with in-depth skills in data analysis, big data technologies, machine learning, and business intelligence tools such as Python, R, SQL, Tableau, and Hadoop. This qualification is designed to prepare individuals for advanced roles in business analytics, enabling them to lead data-driven decision-making processes and optimize business strategies through sophisticated analytical techniques. It is ideal for those seeking to enhance their expertise in the rapidly evolving field of business analytics.

Personal Attributes

Analytical thinking, problem-solving ability, leadership skills, strategic decision-making, technical proficiency, attention to detail, adaptability, strong communication skills, critical thinking, data-driven mindset.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MEP/N1515: Basics of R Programming](#)
2. [MEP/N1516: Database Management System](#)
3. [MEP/N1517: Data Visualization Tools - Tableau](#)
4. [MEP/N0259: Excel for Business Applications](#)
5. [MEP/N0256: Personality Development](#)
6. [MEP/N0422: Entrepreneurship Development & Innovation \(EDI\)](#)
7. [MEP/N1520: Basics of Python](#)
8. [MEP/N1521: Data Warehousing and OLAP](#)
9. [MEP/N1522: Big Data Analytics](#)
10. [MEP/N1523: OJT Mandatory - Business Analytics](#)

Electives(mandatory to select at least one):

Elective 1: Logistics and Supply Chain Analytics

The student/ trainee has to select either Elective 1 - Logistics and Supply Chain Analytics or Elective 2 -

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Marketing & Retail Analytics in semester 3.

1. [MEP/N1518: Logistics and Supply Chain Analytics](#)

Elective 2: Marketing & Retail Analytics

The student/ trainee has to select either Elective 1 - Logistics and Supply Chain Analytics or Elective 2 - Marketing & Retail Analytics in semester 3.

1. [MEP/N1519: Marketing and Retail Analytics](#)

Qualification Pack (QP) Parameters

Sector	Management
Sub-Sector	Office Management
Occupation	Operations
Country	India
NSQF Level	5
Credits	40
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2120.0500
Minimum Educational Qualification & Experience	Previous relevant Qualification of NSQF Level (4.5 (Business Analytics Executive))
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	17/12/2027
NSQF Approval Date	17/12/2024
Version	1.0
Reference code on NQR	QG-05-OA-03432-2024-V1-MEPSC
NQR Version	1

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MEP/N1515: Basics of R Programming

Description

This NOS focuses on the fundamental concepts of R programming, including data manipulation, statistical computing, and visualization, to support business analytics and data-driven decision-making.

Scope

The scope covers the following :

- R Programming Fundamentals
- Basic Data Types, Vectors, Lists, and Matrices
- Arrays, Factors, and Data Frames
- Flow Control & Functions
- Charts & Graphs

Elements and Performance Criteria

R Programming Fundamentals

To be competent, the user/individual on the job must be able to:

- PC1.** Install and configure the R programming environment on various operating systems.
- PC2.** Identify and explain the features of R, including how to execute R scripts and manage projects within R.
- PC3.** Utilize comments, reserved words, and identifiers effectively in R scripts.
- PC4.** Declare and use constants and variables in R, applying appropriate operators (arithmetic, relational, logical, assignment, and miscellaneous) in expressions.
- PC5.** Understand and apply operator precedence rules in complex expressions and work with strings effectively in R.

Basic Data Types, Vectors, Lists, and Matrices

To be competent, the user/individual on the job must be able to:

- PC6.** Define and utilize basic data types in R, including numeric, integer, complex, logical, and character types.
- PC7.** Create, combine, and modify vectors, including accessing, deleting, and sorting vector elements.
- PC8.** Perform vector arithmetic, recycling, and element sorting, demonstrating the ability to read and manipulate vector data.
- PC9.** Create and manipulate lists in R, including accessing, updating, and merging list elements.
- PC10.** Convert lists to vectors and demonstrate list operations effectively.
- PC11.** Create matrices, access matrix elements, and perform matrix arithmetic and operations, including manipulation and sorting of matrix data.

Arrays, Factors, and Data Frames

To be competent, the user/individual on the job must be able to:

- PC12.** Create and manipulate arrays in R, including accessing, modifying, and performing arithmetic operations on array elements.

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- PC13.** Define and manipulate factors, including creating factors, accessing components, and modifying factor levels.
- PC14.** Create and manage data frames in R, including accessing and modifying components, aggregating data, and sorting data within data frames.
- PC15.** Perform operations such as merging, reshaping, sub-setting, and converting data types within data frames.

Flow Control & Functions

To be competent, the user/individual on the job must be able to:

- PC16.** Implement decision-making structures in R, including if statements, nested if-else statements, if-else function, and switch statements.
- PC17.** Use loops effectively, including for loops, while loops, and repeat loops, with appropriate loop control statements like break and next.
- PC18.** Define and call functions in R, including functions without arguments and recursive functions.
- PC19.** Utilize built-in R functions for various operations, including mathematical, character, statistical, and date-time functions, ensuring correct application in different scenarios.

Charts & Graphs

To be competent, the user/individual on the job must be able to:

- PC20.** Create bar charts in R, including plotting bars vertically and horizontally, working with categorical data, and generating grouped and stacked bar charts.
- PC21.** Develop histograms, including simple histograms and those with labels, breaks, and density lines.
- PC22.** Plot line graphs, including simple line graphs and graphs with multiple lines.
- PC23.** Generate pie charts in R, including simple pie charts, pie charts with slice percentages, and 3D pie charts.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the basic syntax, data structures, and functions in R programming.
- KU2.** Recognize the role of R in statistical analysis, data visualization, and machine learning applications.
- KU3.** Explain data manipulation techniques using R packages such as dplyr and tidyr.
- KU4.** Identify best practices for writing efficient R scripts and debugging errors.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate analytical findings effectively using R-generated reports and visualizations.
- GS2.** Apply problem-solving skills to clean, process, and analyze data using R.
- GS3.** Use critical thinking to select appropriate statistical techniques for business insights.
- GS4.** Demonstrate proficiency in RStudio and relevant R libraries for data analysis.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>R Programming Fundamentals</i>	10	2	-	-
PC1. Install and configure the R programming environment on various operating systems.	-	-	-	-
PC2. Identify and explain the features of R, including how to execute R scripts and manage projects within R.	-	-	-	-
PC3. Utilize comments, reserved words, and identifiers effectively in R scripts.	-	-	-	-
PC4. Declare and use constants and variables in R, applying appropriate operators (arithmetic, relational, logical, assignment, and miscellaneous) in expressions.	-	-	-	-
PC5. Understand and apply operator precedence rules in complex expressions and work with strings effectively in R.	-	-	-	-
<i>Basic Data Types, Vectors, Lists, and Matrices</i>	20	2	-	-
PC6. Define and utilize basic data types in R, including numeric, integer, complex, logical, and character types.	-	-	-	-
PC7. Create, combine, and modify vectors, including accessing, deleting, and sorting vector elements.	-	-	-	-
PC8. Perform vector arithmetic, recycling, and element sorting, demonstrating the ability to read and manipulate vector data.	-	-	-	-
PC9. Create and manipulate lists in R, including accessing, updating, and merging list elements.	-	-	-	-
PC10. Convert lists to vectors and demonstrate list operations effectively.	-	-	-	-
PC11. Create matrices, access matrix elements, and perform matrix arithmetic and operations, including manipulation and sorting of matrix data.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Arrays, Factors, and Data Frames</i>	20	2	-	-
PC12. Create and manipulate arrays in R, including accessing, modifying, and performing arithmetic operations on array elements.	-	-	-	-
PC13. Define and manipulate factors, including creating factors, accessing components, and modifying factor levels.	-	-	-	-
PC14. Create and manage data frames in R, including accessing and modifying components, aggregating data, and sorting data within data frames.	-	-	-	-
PC15. Perform operations such as merging, reshaping, sub-setting, and converting data types within data frames.	-	-	-	-
<i>Flow Control & Functions</i>	20	2	-	-
PC16. Implement decision-making structures in R, including if statements, nested if-else statements, if-else function, and switch statements.	-	-	-	-
PC17. Use loops effectively, including for loops, while loops, and repeat loops, with appropriate loop control statements like break and next.	-	-	-	-
PC18. Define and call functions in R, including functions without arguments and recursive functions.	-	-	-	-
PC19. Utilize built-in R functions for various operations, including mathematical, character, statistical, and date-time functions, ensuring correct application in different scenarios.	-	-	-	-
<i>Charts & Graphs</i>	20	2	-	-
PC20. Create bar charts in R, including plotting bars vertically and horizontally, working with categorical data, and generating grouped and stacked bar charts.	-	-	-	-
PC21. Develop histograms, including simple histograms and those with labels, breaks, and density lines.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. Plot line graphs, including simple line graphs and graphs with multiple lines.	-	-	-	-
PC23. Generate pie charts in R, including simple pie charts, pie charts with slice percentages, and 3D pie charts.	-	-	-	-
NOS Total	90	10	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	MEP/N1515
NOS Name	Basics of R Programming
Sector	Management
Sub-Sector	
Occupation	Operations
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024

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MEP/N1516: Database Management System

Description

This NOS focuses on the fundamentals of database management systems (DBMS), including data storage, retrieval, security, and optimization to support efficient business data management.

Scope

The scope covers the following :

- Database Approach
- Working with MS Access
- Working with Structured Query Language (SQL)
- Working with PL/SQL
- Query Processing and Optimization

Elements and Performance Criteria

Database Approach

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the features, advantages, and disadvantages of the database approach compared to traditional data management methods.
- PC2.** Identify and describe the components of a Database Management System (DBMS), including hardware, software, data, procedures, and database access language.
- PC3.** Differentiate between various data models, including hierarchical, network, relational, and Entity-Relationship (ER) models.
- PC4.** Perform ER analysis, defining attributes and domains, and apply integrity constraints such as primary keys, foreign keys, and unique constraints.
- PC5.** Explain the concepts of normalization and apply normalization techniques to design databases in 1NF, 2NF, and 3NF to eliminate redundancy and ensure data integrity.
- PC6.** Maintain data security, confidentiality and follow ethical conduct always while handling different data

Working with MS Access

To be competent, the user/individual on the job must be able to:

- PC7.** Create tables in MS Access, defining data types and field properties according to the requirements of the database.
- PC8.** Establish and manage relationships between tables in MS Access to maintain referential integrity.
- PC9.** Design and create forms in MS Access for efficient data entry and retrieval.
- PC10.** Construct and execute queries in MS Access, including simple queries, cross-tab queries, and queries designed for specific data retrieval needs.
- PC11.** Generate reports in MS Access using report wizard and query design, creating simple reports and cross-tab reports that summarize and present data effectively.

Working with Structured Query Language (SQL)

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To be competent, the user/individual on the job must be able to:

- PC12.** Write and execute Data Definition Language (DDL) statements in SQL to create, drop, and alter database structures.
- PC13.** Perform data manipulation using Data Manipulation Language (DML) statements such as insert, select, delete, and update.
- PC14.** Utilize Oracle functions, apply join conditions, and set operators to retrieve and manipulate data from multiple tables.
- PC15.** Implement the Order By clause in SQL queries to sort data according to specific criteria.

Working with PL/SQL

To be competent, the user/individual on the job must be able to:

- PC16.** Apply control structures in PL/SQL, including loops, conditional statements, and exception handling to develop robust database programs.
- PC17.** Define and use functions and triggers in PL/SQL to automate database operations and maintain data integrity.

Query Processing and Optimization

To be competent, the user/individual on the job must be able to:

- PC18.** Explain the ACID properties of transactions and their importance in ensuring data consistency and reliability.
- PC19.** Manage transaction processing and apply concurrency control techniques to prevent data conflicts in multi-user environments.
- PC20.** Describe the principles of database recovery and apply appropriate techniques to restore databases following a failure, ensuring minimal data loss.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the key concepts of database management systems, including relational and non-relational databases.
- KU2.** Recognize the role of SQL and NoSQL databases in storing and managing structured and unstructured data.
- KU3.** Explain data modeling, normalization, and indexing techniques for optimizing database performance.
- KU4.** Identify best practices for database security, backup, and recovery management.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate database structures and queries effectively to technical and non-technical stakeholders.
- GS2.** Apply analytical skills to design, manage, and optimize databases for business applications.
- GS3.** Use problem-solving techniques to troubleshoot database performance and integrity issues.
- GS4.** Demonstrate proficiency in SQL, database management tools, and cloud-based database solutions.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Database Approach</i>	20	2	-	-
PC1. Explain the features, advantages, and disadvantages of the database approach compared to traditional data management methods.	-	-	-	-
PC2. Identify and describe the components of a Database Management System (DBMS), including hardware, software, data, procedures, and database access language.	-	-	-	-
PC3. Differentiate between various data models, including hierarchical, network, relational, and Entity-Relationship (ER) models.	-	-	-	-
PC4. Perform ER analysis, defining attributes and domains, and apply integrity constraints such as primary keys, foreign keys, and unique constraints.	-	-	-	-
PC5. Explain the concepts of normalization and apply normalization techniques to design databases in 1NF, 2NF, and 3NF to eliminate redundancy and ensure data integrity.	-	-	-	-
PC6. Maintain data security, confidentiality and follow ethical conduct always while handling different data	-	-	-	-
<i>Working with MS Access</i>	10	2	-	-
PC7. Create tables in MS Access, defining data types and field properties according to the requirements of the database.	-	-	-	-
PC8. Establish and manage relationships between tables in MS Access to maintain referential integrity.	-	-	-	-
PC9. Design and create forms in MS Access for efficient data entry and retrieval.	-	-	-	-
PC10. Construct and execute queries in MS Access, including simple queries, cross-tab queries, and queries designed for specific data retrieval needs.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Generate reports in MS Access using report wizard and query design, creating simple reports and cross-tab reports that summarize and present data effectively.	-	-	-	-
<i>Working with Structured Query Language (SQL)</i>	20	2	-	-
PC12. Write and execute Data Definition Language (DDL) statements in SQL to create, drop, and alter database structures.	-	-	-	-
PC13. Perform data manipulation using Data Manipulation Language (DML) statements such as insert, select, delete, and update.	-	-	-	-
PC14. Utilize Oracle functions, apply join conditions, and set operators to retrieve and manipulate data from multiple tables.	-	-	-	-
PC15. Implement the Order By clause in SQL queries to sort data according to specific criteria.	-	-	-	-
<i>Working with PL/SQL</i>	20	2	-	-
PC16. Apply control structures in PL/SQL, including loops, conditional statements, and exception handling to develop robust database programs.	-	-	-	-
PC17. Define and use functions and triggers in PL/SQL to automate database operations and maintain data integrity.	-	-	-	-
<i>Query Processing and Optimization</i>	20	2	-	-
PC18. Explain the ACID properties of transactions and their importance in ensuring data consistency and reliability.	-	-	-	-
PC19. Manage transaction processing and apply concurrency control techniques to prevent data conflicts in multi-user environments.	-	-	-	-
PC20. Describe the principles of database recovery and apply appropriate techniques to restore databases following a failure, ensuring minimal data loss.	-	-	-	-
NOS Total	90	10	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	MEP/N1516
NOS Name	Database Management System
Sector	Management
Sub-Sector	
Occupation	Operations
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024

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MEP/N1517: Data Visualization Tools - Tableau

Description

This NOS focuses on using Tableau for data visualization, analysis, and reporting to enhance business intelligence and support data-driven decision-making.

Scope

The scope covers the following :

- Tableau Basic Reports
- Tableau Charts
- HR analytics tools and techniques
- Tableau Advanced Reports
- Tableau Calculations & Filters

Elements and Performance Criteria

Tableau Basic Reports

To be competent, the user/individual on the job must be able to:

- PC1.** Connect to Excel files within Tableau, importing data for analysis.
- PC2.** Connect to text files and import data efficiently into Tableau for report generation.
- PC3.** Integrate data from Microsoft Analysis Services with Tableau, ensuring correct data interpretation.
- PC4.** Create and remove hierarchies in Tableau to organize data and enable drill-down analysis.
- PC5.** Define and implement parameters in Tableau to control data visualizations dynamically.
- PC6.** Utilize grouping techniques to categorize data, applying examples to create meaningful data segments.
- PC7.** Design your first report in Tableau, applying data labels for clarity.
- PC8.** Organize data using folders and apply sorting techniques to enhance data visualization.
- PC9.** Combine data sets in Tableau, creating comprehensive reports that provide a holistic view of the data.

Tableau Charts

To be competent, the user/individual on the job must be able to:

- PC10.** Create various chart types in Tableau, including area, bar, box, bubble, and bump charts, to represent data visually.
- PC11.** Develop more complex visualizations such as bullet, circle, Gantt, line, and lollipop charts.
- PC12.** Utilize advanced chart types like Pareto, pie, scatter, stacked, tree, world cloud, and waterfall charts to provide in-depth data analysis.
- PC13.** Create filled maps, crosstab, combined, motion, and heatmap visualizations to represent geographical and temporal data.

HR analytics tools and techniques

To be competent, the user/individual on the job must be able to:

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- PC14.** Discuss the importance of data in HR analytics.
- PC15.** Identify the types of data and data capturing methods.
- PC16.** Explain the techniques for data examination and purification.
- PC17.** Describe various data analyzing techniques used in HR analytics.

Tableau Advanced Reports

To be competent, the user/individual on the job must be able to:

- PC18.** Create dual-axis reports in Tableau, effectively visualizing multiple data sets on a single chart.
- PC19.** Implement blended and individual axis reports to compare data across different measures.
- PC20.** Add reference lines and reference distributions to reports, providing additional context and insights.
- PC21.** Create and customize basic map visualizations, integrating Mapbox maps as background maps for enhanced geographical analysis.

Tableau Calculations & Filters

To be competent, the user/individual on the job must be able to:

- PC22.** Create calculated fields in Tableau to perform custom calculations and derive new metrics.
- PC23.** Apply basic approaches to calculate ranks within data sets, enabling comparative analysis.
- PC24.** Utilize top and bottom filters to focus on key data points.
- PC25.** Implement context and slicing filters to refine data visualizations.
- PC26.** Apply data source filters, extract filters, and other filtering techniques to manage large data sets and enhance report performance.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the fundamentals of data visualization and its significance in business analytics.
- KU2.** Recognize the key features and functionalities of Tableau, including dashboards, charts, and data connections.
- KU3.** Explain techniques to transform raw data into meaningful insights using Tableau's visualization tools
- KU4.** Identify best practices for designing interactive and user-friendly reports to improve decision-making.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate insights effectively using Tableau dashboards and reports.
- GS2.** Apply analytical skills to interpret data trends and patterns for business strategy.
- GS3.** Use problem-solving techniques to optimize data representation and enhance report accuracy.
- GS4.** Demonstrate proficiency in Tableau tools, including calculated fields, parameters, and data blending.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Tableau Basic Reports</i>	10	4	-	-
PC1. Connect to Excel files within Tableau, importing data for analysis.	-	-	-	-
PC2. Connect to text files and import data efficiently into Tableau for report generation.	-	-	-	-
PC3. Integrate data from Microsoft Analysis Services with Tableau, ensuring correct data interpretation.	-	-	-	-
PC4. Create and remove hierarchies in Tableau to organize data and enable drill-down analysis.	-	-	-	-
PC5. Define and implement parameters in Tableau to control data visualizations dynamically.	-	-	-	-
PC6. Utilize grouping techniques to categorize data, applying examples to create meaningful data segments.	-	-	-	-
PC7. Design your first report in Tableau, applying data labels for clarity.	-	-	-	-
PC8. Organize data using folders and apply sorting techniques to enhance data visualization.	-	-	-	-
PC9. Combine data sets in Tableau, creating comprehensive reports that provide a holistic view of the data.	-	-	-	-
<i>Tableau Charts</i>	20	4	-	-
PC10. Create various chart types in Tableau, including area, bar, box, bubble, and bump charts, to represent data visually.	-	-	-	-
PC11. Develop more complex visualizations such as bullet, circle, Gantt, line, and lollipop charts.	-	-	-	-
PC12. Utilize advanced chart types like Pareto, pie, scatter, stacked, tree, world cloud, and waterfall charts to provide in-depth data analysis.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Create filled maps, crosstab, combined, motion, and heatmap visualizations to represent geographical and temporal data.	-	-	-	-
<i>HR analytics tools and techniques</i>	20	4	-	-
PC14. Discuss the importance of data in HR analytics.	-	-	-	-
PC15. Identify the types of data and data capturing methods.	-	-	-	-
PC16. Explain the techniques for data examination and purification.	-	-	-	-
PC17. Describe various data analyzing techniques used in HR analytics.	-	-	-	-
<i>Tableau Advanced Reports</i>	10	4	-	-
PC18. Create dual-axis reports in Tableau, effectively visualizing multiple data sets on a single chart.	-	-	-	-
PC19. Implement blended and individual axis reports to compare data across different measures.	-	-	-	-
PC20. Add reference lines and reference distributions to reports, providing additional context and insights.	-	-	-	-
PC21. Create and customize basic map visualizations, integrating Mapbox maps as background maps for enhanced geographical analysis.	-	-	-	-
<i>Tableau Calculations & Filters</i>	20	4	-	-
PC22. Create calculated fields in Tableau to perform custom calculations and derive new metrics.	-	-	-	-
PC23. Apply basic approaches to calculate ranks within data sets, enabling comparative analysis.	-	-	-	-
PC24. Utilize top and bottom filters to focus on key data points.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC25. Implement context and slicing filters to refine data visualizations.	-	-	-	-
PC26. Apply data source filters, extract filters, and other filtering techniques to manage large data sets and enhance report performance.	-	-	-	-
NOS Total	80	20	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	MEP/N1517
NOS Name	Data Visualization Tools - Tableau
Sector	Management
Sub-Sector	
Occupation	Operations
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024

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MEP/N0259: Excel for Business Applications

Description

This NOS focuses on using Microsoft Excel for business applications, including data management, analysis, reporting, and automation to enhance productivity and decision-making.

Scope

The scope covers the following :

- Introduction to Excel
- Data Management and Complex Functions in Excel
- Conditional Formatting, Data Validation, and Advanced Referencing
- Advanced Data Analysis, PivotTables, and Macros
- Collaboration, Security, and Workbook Management in Excel

Elements and Performance Criteria

Introduction to Excel

To be competent, the user/individual on the job must be able to:

- PC1.** Enter and edit data in Excel, applying basic formulae and functions to perform calculations.
- PC2.** Insert and delete rows/columns, and adjust cell formatting, including fonts, colours, and borders.
- PC3.** Merge and align cells, ensuring data is presented clearly and accurately.
- PC4.** Use page break preview and page layout view to prepare worksheets for printing, including setting headers and footers.
- PC5.** Create and format charts quickly by selecting data and applying basic chart types.
- PC6.** Create and format basic tables, using table styles, calculated columns, and applying sorting and filtering.

Data Management and Complex Functions in Excel

To be competent, the user/individual on the job must be able to:

- PC7.** Create, sort, and filter lists/tables of data using range names for efficient data management.
- PC8.** Apply custom number formats, including scaling numbers and formatting dates and times, to improve data readability.
- PC9.** Use mathematical, text, logical, and reference functions to perform advanced data analysis in Excel.

Conditional Formatting, Data Validation, and Advanced Referencing

To be competent, the user/individual on the job must be able to:

- PC10.** Create and apply cell rules for conditional formatting, using data bars and colour sets to highlight key data points.
- PC11.** Implement styles and themes, creating custom styles to maintain consistency in worksheet formatting.
- PC12.** Set cell validation and protection, including grouping and outlining data, and adding cell comments for clarity.

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PC13. Utilize absolute references and range names to fix row/column references in formulas and label ranges automatically.

Advanced Data Analysis, PivotTables, and Macros

To be competent, the user/individual on the job must be able to:

PC14. Define alternative data sets and use Goal Seek and Solver to find optimal solutions for business scenarios.

PC15. Analyse data dynamically using PivotTables, including filtering, editing, and formatting PivotTable data, and creating PivotTables from external data sources.

PC16. Create and modify macros to automate repetitive tasks in Excel, and run macros using buttons.

PC17. Create shapes and mathematical equations to enhance the visual presentation of data in Excel.

Collaboration, Security, and Workbook Management in Excel

To be competent, the user/individual on the job must be able to:

PC18. Create hyperlinks and paste charts into other documents to integrate Excel data with other applications.

PC19. Share workbooks, manage comments, track changes, and protect workbooks/worksheets to facilitate collaboration with colleagues.

PC20. Authenticate workbooks to ensure data integrity and save workbooks for the web for broader accessibility.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the fundamental and advanced functions of Excel for business applications.

KU2. Recognize the use of formulas, functions, and data validation techniques for accurate calculations.

KU3. Explain data visualization techniques, including charts, PivotTables, and dashboards, for business reporting.

KU4. Identify automation tools such as Macros and VBA to streamline business processes.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Communicate business insights effectively using Excel-based reports and presentations.

GS2. Apply analytical skills to interpret and manipulate business data for decision-making.

GS3. Use problem-solving techniques to optimize data management and workflow automation.

GS4. Demonstrate proficiency in Excel tools to improve efficiency in business operations.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Excel</i>	10	2	-	-
PC1. Enter and edit data in Excel, applying basic formulae and functions to perform calculations.	-	-	-	-
PC2. Insert and delete rows/columns, and adjust cell formatting, including fonts, colours, and borders.	-	-	-	-
PC3. Merge and align cells, ensuring data is presented clearly and accurately.	-	-	-	-
PC4. Use page break preview and page layout view to prepare worksheets for printing, including setting headers and footers.	-	-	-	-
PC5. Create and format charts quickly by selecting data and applying basic chart types.	-	-	-	-
PC6. Create and format basic tables, using table styles, calculated columns, and applying sorting and filtering.	-	-	-	-
<i>Data Management and Complex Functions in Excel</i>	20	2	-	-
PC7. Create, sort, and filter lists/tables of data using range names for efficient data management.	-	-	-	-
PC8. Apply custom number formats, including scaling numbers and formatting dates and times, to improve data readability.	-	-	-	-
PC9. Use mathematical, text, logical, and reference functions to perform advanced data analysis in Excel.	-	-	-	-
<i>Conditional Formatting, Data Validation, and Advanced Referencing</i>	20	2	-	-
PC10. Create and apply cell rules for conditional formatting, using data bars and colour sets to highlight key data points.	-	-	-	-
PC11. Implement styles and themes, creating custom styles to maintain consistency in worksheet formatting.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Set cell validation and protection, including grouping and outlining data, and adding cell comments for clarity.	-	-	-	-
PC13. Utilize absolute references and range names to fix row/column references in formulas and label ranges automatically.	-	-	-	-
<i>Advanced Data Analysis, PivotTables, and Macros</i>	20	2	-	-
PC14. Define alternative data sets and use Goal Seek and Solver to find optimal solutions for business scenarios.	-	-	-	-
PC15. Analyse data dynamically using PivotTables, including filtering, editing, and formatting PivotTable data, and creating PivotTables from external data sources.	-	-	-	-
PC16. Create and modify macros to automate repetitive tasks in Excel, and run macros using buttons.	-	-	-	-
PC17. Create shapes and mathematical equations to enhance the visual presentation of data in Excel.	-	-	-	-
<i>Collaboration, Security, and Workbook Management in Excel</i>	20	2	-	-
PC18. Create hyperlinks and paste charts into other documents to integrate Excel data with other applications.	-	-	-	-
PC19. Share workbooks, manage comments, track changes, and protect workbooks/worksheets to facilitate collaboration with colleagues.	-	-	-	-
PC20. Authenticate workbooks to ensure data integrity and save workbooks for the web for broader accessibility.	-	-	-	-
NOS Total	90	10	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MEP/N0259
NOS Name	Excel for Business Applications
Sector	Management
Sub-Sector	
Occupation	Office Support
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024

Qualification Pack

MEP/N0256: Personality Development

Description

This NOS focuses on enhancing personal and professional attributes, including communication skills, confidence, emotional intelligence, and workplace etiquette, to improve overall effectiveness and career growth.

Scope

The scope covers the following :

- Interview skills and techniques
- Group Discussion
- Presentation skills
- Conversational Skills
- Team Building
- Writing Skills for Managers

Elements and Performance Criteria

Interview skills and techniques

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the purpose of interviews in the context of recruitment and selection.
- PC2.** Identify the basic components and structure of an effective interview.
- PC3.** Describe various types of interviews (e.g., structured, unstructured, behavioral).
- PC4.** Compare and contrast the different interview types and their applications.
- PC5.** Demonstrate an effective self-introduction in an interview setting.
- PC6.** Provide constructive feedback to peers on their interview introductions.
- PC7.** Outline the key steps in preparing for an interview.
- PC8.** Create a checklist of essential preparations for an interview.
- PC9.** Explain the guidelines for conducting successful telephonic interviews.
- PC10.** Discuss the best practices for personal interviews.
- PC11.** Introduce the concept and purpose of group interviews.
- PC12.** Participate in a mock group interview and provide feedback on the experience.

Group Discussion

To be competent, the user/individual on the job must be able to:

- PC13.** List key points to remember for effective participation in group discussions.
- PC14.** Identify steps to succeed in group discussions and common do's and don'ts.
- PC15.** Engage in practice group discussions on current topics and provide constructive feedback.

Presentation skills

To be competent, the user/individual on the job must be able to:

- PC16.** Explain the importance of planning and preparation for a successful presentation.

Qualification Pack

- PC17.** Develop a comprehensive presentation plan.
- PC18.** Describe key tips for creating effective PowerPoint presentations, including structure and content.
- PC19.** Create and deliver a PowerPoint presentation, focusing on introduction, middle, and closing.

Conversational Skills

To be competent, the user/individual on the job must be able to:

- PC20.** Define the purpose and importance of effective conversations in personal and professional contexts.
- PC21.** Identify different types of conversations and their objectives.
- PC22.** List practical tips for improving conversational skills.
- PC23.** Practice implementing these tips in various conversational scenarios.
- PC24.** Demonstrate the ability to ask appropriate questions in a conversation.
- PC25.** Exhibit care, courtesy, and consideration in conversational interactions.
- PC26.** Explain guidelines for effective telephonic conversations, both personal and professional.
- PC27.** Practice conversational skills with strangers in simulated scenarios.

Team Building

To be competent, the user/individual on the job must be able to:

- PC28.** Describe the stages of team development and effective team management strategies.
- PC29.** Identify the roles and responsibilities of team members in achieving team goals.
- PC30.** List the characteristics of a successful team.
- PC31.** Analyze the dynamics of a high-performing team.
- PC32.** Participate in team-building activities to enhance team cohesion and collaboration.
- PC33.** Reflect on the outcomes of team-building exercises and their impact on team performance.
- PC34.** Explain the importance of team building in achieving organizational objectives.
- PC35.** Develop strategies to foster team spirit and unity.

Writing Skills for Managers

To be competent, the user/individual on the job must be able to:

- PC36.** Demonstrate effective email writing skills for personal and professional communication.
- PC37.** Identify the key components of a well-structured email.
- PC38.** Explain the benefits of diary writing for personal and professional development.
- PC39.** Develop a habit of regular diary writing.
- PC40.** Implement strategies to enhance vocabulary.
- PC41.** Apply newly learned vocabulary in written and spoken communication.
- PC42.** Write clear and concise objectives for resumes.
- PC43.** Tailor resume objectives to align with specific job roles.
- PC44.** Explain the key elements of effective report writing.
- PC45.** Write comprehensive and well-structured reports.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** Understand the key elements of personality development, including self-awareness, confidence, and emotional intelligence.
- KU2.** Recognize the importance of effective communication, interpersonal skills, and professional etiquette.
- KU3.** Explain strategies for personal grooming, time management, and stress management in a professional setting.
- KU4.** Identify techniques for developing leadership qualities and a positive mindset.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate effectively in professional and social environments.
- GS2.** Apply critical thinking to enhance problem-solving and decision-making skills.
- GS3.** Demonstrate adaptability and resilience in handling workplace challenges.
- GS4.** Use self-management techniques to improve confidence, productivity, and personal growth.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Interview skills and techniques</i>	10	4	-	-
PC1. Explain the purpose of interviews in the context of recruitment and selection.	-	-	-	-
PC2. Identify the basic components and structure of an effective interview.	-	-	-	-
PC3. Describe various types of interviews (e.g., structured, unstructured, behavioral).	-	-	-	-
PC4. Compare and contrast the different interview types and their applications.	-	-	-	-
PC5. Demonstrate an effective self-introduction in an interview setting.	-	-	-	-
PC6. Provide constructive feedback to peers on their interview introductions.	-	-	-	-
PC7. Outline the key steps in preparing for an interview.	-	-	-	-
PC8. Create a checklist of essential preparations for an interview.	-	-	-	-
PC9. Explain the guidelines for conducting successful telephonic interviews.	-	-	-	-
PC10. Discuss the best practices for personal interviews.	-	-	-	-
PC11. Introduce the concept and purpose of group interviews.	-	-	-	-
PC12. Participate in a mock group interview and provide feedback on the experience.	-	-	-	-
<i>Group Discussion</i>	20	4	-	-
PC13. List key points to remember for effective participation in group discussions.	-	-	-	-
PC14. Identify steps to succeed in group discussions and common do's and don'ts.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Engage in practice group discussions on current topics and provide constructive feedback.	-	-	-	-
<i>Presentation skills</i>	20	4	-	-
PC16. Explain the importance of planning and preparation for a successful presentation.	-	-	-	-
PC17. Develop a comprehensive presentation plan.	-	-	-	-
PC18. Describe key tips for creating effective PowerPoint presentations, including structure and content.	-	-	-	-
PC19. Create and deliver a PowerPoint presentation, focusing on introduction, middle, and closing.	-	-	-	-
<i>Conversational Skills</i>	10	4	-	-
PC20. Define the purpose and importance of effective conversations in personal and professional contexts.	-	-	-	-
PC21. Identify different types of conversations and their objectives.	-	-	-	-
PC22. List practical tips for improving conversational skills.	-	-	-	-
PC23. Practice implementing these tips in various conversational scenarios.	-	-	-	-
PC24. Demonstrate the ability to ask appropriate questions in a conversation.	-	-	-	-
PC25. Exhibit care, courtesy, and consideration in conversational interactions.	-	-	-	-
PC26. Explain guidelines for effective telephonic conversations, both personal and professional.	-	-	-	-
PC27. Practice conversational skills with strangers in simulated scenarios.	-	-	-	-
<i>Team Building</i>	10	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. Describe the stages of team development and effective team management strategies.	-	-	-	-
PC29. Identify the roles and responsibilities of team members in achieving team goals.	-	-	-	-
PC30. List the characteristics of a successful team.	-	-	-	-
PC31. Analyze the dynamics of a high-performing team.	-	-	-	-
PC32. Participate in team-building activities to enhance team cohesion and collaboration.	-	-	-	-
PC33. Reflect on the outcomes of team-building exercises and their impact on team performance.	-	-	-	-
PC34. Explain the importance of team building in achieving organizational objectives.	-	-	-	-
PC35. Develop strategies to foster team spirit and unity.	-	-	-	-
<i>Writing Skills for Managers</i>	10	2	-	-
PC36. Demonstrate effective email writing skills for personal and professional communication.	-	-	-	-
PC37. Identify the key components of a well-structured email.	-	-	-	-
PC38. Explain the benefits of diary writing for personal and professional development.	-	-	-	-
PC39. Develop a habit of regular diary writing.	-	-	-	-
PC40. Implement strategies to enhance vocabulary.	-	-	-	-
PC41. Apply newly learned vocabulary in written and spoken communication.	-	-	-	-
PC42. Write clear and concise objectives for resumes.	-	-	-	-
PC43. Tailor resume objectives to align with specific job roles.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC44. Explain the key elements of effective report writing.	-	-	-	-
PC45. Write comprehensive and well-structured reports.	-	-	-	-
NOS Total	80	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MEP/N0256
NOS Name	Personality Development
Sector	Management
Sub-Sector	
Occupation	Office Support
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024

Qualification Pack

MEP/N0422: Entrepreneurship Development & Innovation (EDI)

Description

The unit describes searching for business ideas, business planning, legal and regulatory considerations and financial management for running an enterprise.

Scope

The scope covers the following :

- Introduction to Entrepreneurship
- Identifying Business Opportunities
- Business Planning and Strategy
- Legal and Regulatory Considerations
- Financial Management for Entrepreneurs

Elements and Performance Criteria

Introduction to Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC1.** Define the concept of entrepreneurship and its significance in driving economic growth and innovation
- PC2.** Identify key characteristics and traits of successful entrepreneurs, including creativity, resilience, and risk-taking propensity
- PC3.** Analyse the role of entrepreneurship in creating job opportunities, fostering technological advancements, and addressing societal needs
- PC4.** Understand the difference between small business entrepreneurship, social entrepreneurship, and scalable startup ventures
- PC5.** Cultivate an entrepreneurial mindset by embracing uncertainty, learning from failures, and seeking opportunities for innovation and growth
- PC6.** Explore real-world examples of successful entrepreneurs and their entrepreneurial journeys across different industries and sectors
- PC7.** Discuss the importance of entrepreneurship education and skill development in nurturing the next generation of innovators and business leaders
- PC8.** Evaluate the impact of entrepreneurial ecosystems, including incubators, accelerators, and coworking spaces, in supporting startup ventures
- PC9.** Reflect on personal entrepreneurial aspirations, goals, and motivations, and develop a preliminary understanding of potential business ideas

Identifying Business Opportunities

To be competent, the user/individual on the job must be able to:

- PC10.** Utilise techniques for idea generation and opportunity recognition, such as brainstorming, problem-solving, and trend analysis
- PC11.** Conduct market research to identify unmet customer needs, emerging trends, and potential business opportunities

Qualification Pack

- PC12.** Evaluate the feasibility and viability of business ideas through market validation, prototype testing, and customer feedback
- PC13.** Analyse industry trends, competitive landscapes, and market dynamics to identify niche markets and competitive advantages
- PC14.** Segment target customer groups based on demographics, psychographics, and behaviour to tailor products or services to specific market segments
- PC15.** Identify potential barriers to entry, regulatory challenges, and risks associated with pursuing specific business opportunities
- PC16.** Develop a comprehensive understanding of the customer journey and identify touchpoints for delivering value to customers
- PC17.** Explore various business models and revenue streams to monetize business ideas effectively
- PC18.** Collaborate with industry experts, mentors, and advisors to gain insights and perspectives on potential business opportunities
- PC19.** Synthesize findings from opportunity identification processes to refine business ideas and develop actionable plans for further exploration and validation

Business Planning and Strategy

To be competent, the user/individual on the job must be able to:

- PC20.** Develop a clear and concise business plan outlining the vision, mission, and goals of the venture
- PC21.** Identify target markets, customer segments, and value propositions to differentiate the business from competitors
- PC22.** Conduct a comprehensive market analysis to assess market size, growth potential, and competitive landscape
- PC23.** Formulate marketing strategies and tactics to reach target customers and promote brand awareness
- PC24.** Outline operational plans, including production processes, supply chain management, and resource allocation
- PC25.** Develop financial projections, including revenue forecasts, expense budgets, and cash flow statements
- PC26.** Evaluate scalability and growth opportunities for the business and develop strategies for expansion
- PC27.** Identify potential risks and challenges associated with the business venture and develop contingency plans
- PC28.** Collaborate with team members, advisors, and stakeholders to refine and finalize the business plan
- PC29.** Present the business plan effectively to potential investors, partners, and stakeholders to secure funding and support for the venture

Legal and Regulatory Considerations

To be competent, the user/individual on the job must be able to:

- PC30.** Understand different legal structures for businesses, including sole proprietorship, partnership, corporation, and limited liability company (LLC)
- PC31.** Navigate the process of registering a business entity and obtaining necessary licenses and permits

Qualification Pack

- PC32.** Identify and protect intellectual property assets, including patents, trademarks, copyrights, and trade secrets
- PC33.** Comply with taxation laws and regulations, including income tax, sales tax, and employment taxes
- PC34.** Draft and negotiate contracts and agreements with suppliers, customers, and other business partners
- PC35.** Understand employer obligations and labour laws related to hiring employees, wages, benefits, and workplace safety.
- PC36.** Manage legal risks and liabilities associated with business operations, including product liability and professional indemnity
- PC37.** Address regulatory requirements specific to the industry or sector in which the business operates
- PC38.** Establish policies and procedures for data privacy and security compliance in accordance with relevant regulations
- PC39.** Seek legal advice and consultation from qualified professionals to ensure compliance with all legal and regulatory requirements

Financial Management for Entrepreneurs

To be competent, the user/individual on the job must be able to:

- PC40.** Develop a basic understanding of financial management principles, including budgeting, forecasting, and cash flow management
- PC41.** Identify sources of funding for startup ventures, including personal savings, loans, grants, angel investors, and venture capital
- PC42.** Evaluate the pros and cons of different funding options and determine the most suitable financing strategy for the business
- PC43.** Evaluate financial statements, including income statements, balance sheets, and cash flow statements, to track business performance
- PC44.** Interpret financial ratios and metrics to assess the financial health and profitability of the business
- PC45.** Implement strategies for cost control, expense management, and profit maximization
- PC46.** Monitor cash flow and working capital to ensure liquidity and solvency of the business
- PC47.** Prepare financial forecasts and projections to support business planning and decision-making processes
- PC48.** Manage financial risks, including market risks, credit risks, and liquidity risks, through effective risk management strategies
- PC49.** Utilize financial management tools and software to streamline financial processes and improve decision-making efficiency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Principles and concepts of marketing management and their application in digital and traditional marketing
- KU2.** Digital marketing channels, including SEO, SEM, social media, email, and content marketing

Qualification Pack

- KU3.** Web design, content management systems (CMS), and web analytics
- KU4.** Creating and managing online presence
- KU5.** Marketing strategies, including mobile marketing and ecommerce
- KU6.** Consumer behaviour and purchasing decisions
- KU7.** Ways of advertising to contribute to brand awareness and sales generation
- KU8.** Importance of brand identity
- KU9.** Communication strategies, and advertising in shaping consumer perceptions
- KU10.** Generative AI and automation tools
- KU11.** Personalized marketing approaches
- KU12.** Significance of marketing analytics in tracking performance, measuring ROI, and optimizing campaigns
- KU13.** Principles of business communication, both verbal and written, for effective professional interactions
- KU14.** Entrepreneurial mindset, innovation, and how businesses develop and adapt to market trends
- KU15.** Importance of employability skills such as teamwork, adaptability, and professionalism in the workplace
- KU16.** Impact of environmental sustainability and ethical practices in modern business and marketing strategies
- KU17.** Customer relationship management (CRM) systems and their role in managing customer data
- KU18.** Role of data privacy and security in marketing, particularly with the use of AI and data-driven strategies
- KU19.** Significance of aligning marketing efforts with overall business goals to drive growth and profitability
- KU20.** Use of various digital marketing tools and platforms for campaign management and performance tracking
- KU21.** Importance of customer engagement and retention strategies in both online and offline marketing efforts
- KU22.** Fundamentals of data-driven decision-making and how data analysis informs marketing strategies
- KU23.** How marketing campaigns are structured, from planning and budgeting to execution and evaluation
- KU24.** Role of consumer behaviour insights in designing targeted marketing campaigns that meet specific needs and preferences
- KU25.** Ethical considerations in digital marketing, including data usage, transparency, and consumer protection
- KU26.** Concept of integrated marketing communications (IMC) and how different marketing channels work together to achieve business objectives

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** Write factual and quantitative information correctly
- GS2.** Read and extract relevant information from documentation including purpose, nature of document, etc
- GS3.** Communicate effectively with co-workers and clients
- GS4.** Use effective listening and probing/questioning skills
- GS5.** Apply digital literacy and use digital skills securely
- GS6.** Plan day to day tasks to achieve optimum productivity
- GS7.** Manage distractions
- GS8.** Be patient and courteous under all circumstances and situations
- GS9.** Apply problem solving skills that require negotiation to resolve problems of a difficult nature
- GS10.** Breakdown relevant work process into its constituent activities for ease of analysis
- GS11.** Estimate time and effort required to complete a task accurately
- GS12.** Seek clarification on problems and its solution from others

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Entrepreneurship</i>	20	-	-	-
PC1. Define the concept of entrepreneurship and its significance in driving economic growth and innovation	-	-	-	-
PC2. Identify key characteristics and traits of successful entrepreneurs, including creativity, resilience, and risk-taking propensity	-	-	-	-
PC3. Analyse the role of entrepreneurship in creating job opportunities, fostering technological advancements, and addressing societal needs	-	-	-	-
PC4. Understand the difference between small business entrepreneurship, social entrepreneurship, and scalable startup ventures	-	-	-	-
PC5. Cultivate an entrepreneurial mindset by embracing uncertainty, learning from failures, and seeking opportunities for innovation and growth	-	-	-	-
PC6. Explore real-world examples of successful entrepreneurs and their entrepreneurial journeys across different industries and sectors	-	-	-	-
PC7. Discuss the importance of entrepreneurship education and skill development in nurturing the next generation of innovators and business leaders	-	-	-	-
PC8. Evaluate the impact of entrepreneurial ecosystems, including incubators, accelerators, and coworking spaces, in supporting startup ventures	-	-	-	-
PC9. Reflect on personal entrepreneurial aspirations, goals, and motivations, and develop a preliminary understanding of potential business ideas	-	-	-	-
<i>Identifying Business Opportunities</i>	20	-	-	-
PC10. Utilise techniques for idea generation and opportunity recognition, such as brainstorming, problem-solving, and trend analysis	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Conduct market research to identify unmet customer needs, emerging trends, and potential business opportunities	-	-	-	-
PC12. Evaluate the feasibility and viability of business ideas through market validation, prototype testing, and customer feedback	-	-	-	-
PC13. Analyse industry trends, competitive landscapes, and market dynamics to identify niche markets and competitive advantages	-	-	-	-
PC14. Segment target customer groups based on demographics, psychographics, and behaviour to tailor products or services to specific market segments	-	-	-	-
PC15. Identify potential barriers to entry, regulatory challenges, and risks associated with pursuing specific business opportunities	-	-	-	-
PC16. Develop a comprehensive understanding of the customer journey and identify touchpoints for delivering value to customers	-	-	-	-
PC17. Explore various business models and revenue streams to monetize business ideas effectively	-	-	-	-
PC18. Collaborate with industry experts, mentors, and advisors to gain insights and perspectives on potential business opportunities	-	-	-	-
PC19. Synthesize findings from opportunity identification processes to refine business ideas and develop actionable plans for further exploration and validation	-	-	-	-
<i>Business Planning and Strategy</i>	20	-	-	-
PC20. Develop a clear and concise business plan outlining the vision, mission, and goals of the venture	-	-	-	-
PC21. Identify target markets, customer segments, and value propositions to differentiate the business from competitors	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. Conduct a comprehensive market analysis to assess market size, growth potential, and competitive landscape	-	-	-	-
PC23. Formulate marketing strategies and tactics to reach target customers and promote brand awareness	-	-	-	-
PC24. Outline operational plans, including production processes, supply chain management, and resource allocation	-	-	-	-
PC25. Develop financial projections, including revenue forecasts, expense budgets, and cash flow statements	-	-	-	-
PC26. Evaluate scalability and growth opportunities for the business and develop strategies for expansion	-	-	-	-
PC27. Identify potential risks and challenges associated with the business venture and develop contingency plans	-	-	-	-
PC28. Collaborate with team members, advisors, and stakeholders to refine and finalize the business plan	-	-	-	-
PC29. Present the business plan effectively to potential investors, partners, and stakeholders to secure funding and support for the venture	-	-	-	-
<i>Legal and Regulatory Considerations</i>	20	-	-	-
PC30. Understand different legal structures for businesses, including sole proprietorship, partnership, corporation, and limited liability company (LLC)	-	-	-	-
PC31. Navigate the process of registering a business entity and obtaining necessary licenses and permits	-	-	-	-
PC32. Identify and protect intellectual property assets, including patents, trademarks, copyrights, and trade secrets	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC33. Comply with taxation laws and regulations, including income tax, sales tax, and employment taxes	-	-	-	-
PC34. Draft and negotiate contracts and agreements with suppliers, customers, and other business partners	-	-	-	-
PC35. Understand employer obligations and labour laws related to hiring employees, wages, benefits, and workplace safety.	-	-	-	-
PC36. Manage legal risks and liabilities associated with business operations, including product liability and professional indemnity	-	-	-	-
PC37. Address regulatory requirements specific to the industry or sector in which the business operates	-	-	-	-
PC38. Establish policies and procedures for data privacy and security compliance in accordance with relevant regulations	-	-	-	-
PC39. Seek legal advice and consultation from qualified professionals to ensure compliance with all legal and regulatory requirements	-	-	-	-
<i>Financial Management for Entrepreneurs</i>	20	-	-	-
PC40. Develop a basic understanding of financial management principles, including budgeting, forecasting, and cash flow management	-	-	-	-
PC41. Identify sources of funding for startup ventures, including personal savings, loans, grants, angel investors, and venture capital	-	-	-	-
PC42. Evaluate the pros and cons of different funding options and determine the most suitable financing strategy for the business	-	-	-	-
PC43. Evaluate financial statements, including income statements, balance sheets, and cash flow statements, to track business performance	-	-	-	-
PC44. Interpret financial ratios and metrics to assess the financial health and profitability of the business	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC45. Implement strategies for cost control, expense management, and profit maximization	-	-	-	-
PC46. Monitor cash flow and working capital to ensure liquidity and solvency of the business	-	-	-	-
PC47. Prepare financial forecasts and projections to support business planning and decision-making processes	-	-	-	-
PC48. Manage financial risks, including market risks, credit risks, and liquidity risks, through effective risk management strategies	-	-	-	-
PC49. Utilize financial management tools and software to streamline financial processes and improve decision-making efficiency	-	-	-	-
NOS Total	100	-	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MEP/N0422
NOS Name	Entrepreneurship Development & Innovation (EDI)
Sector	Management
Sub-Sector	
Occupation	Marketing
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024

Qualification Pack

MEP/N1520: Basics of Python

Description

This NOS focuses on the fundamental concepts of Python programming, including syntax, data structures, and basic scripting, to support data analysis and automation in business applications.

Scope

The scope covers the following :

- Introduction to Python
- Control Statements
- Strings and Text Files
- Design with Classes
- Graphical User Interfaces (GUI)

Elements and Performance Criteria

Introduction to Python

To be competent, the user/individual on the job must be able to:

- PC1.** Install Python on various operating systems and set up the programming environment.
- PC2.** Explain how a program works and describe the program development cycle.
- PC3.** Demonstrate basic input, processing, and output operations using Python, including the use of the print function and reading input from the keyboard.
- PC4.** Implement basic calculations and expressions, utilizing operators and type conversions effectively.
- PC5.** Apply decision structures using if, if-else, and nested decision-making in Python.
- PC6.** Use logical operators and Boolean variables to perform conditional checks.
- PC7.** Implement repetition structures using while and for loops, and demonstrate the use of loops for running totals and input validation.
- PC8.** Work with strings, numeric data types, and character sets, utilizing functions and modules to manipulate and process data.

Control Statements

To be competent, the user/individual on the job must be able to:

- PC9.** Implement definite and conditional iteration in Python programs, formatting text for output appropriately.
- PC10.** Handle file input and output operations, including processing records and managing exceptions during file handling.
- PC11.** Define and call functions, including value-returning functions, and demonstrate the use of local and global variables.
- PC12.** Utilize Python's math module and random number generation within custom functions.
- PC13.** Organize code by storing functions in modules, ensuring modular design and reusability.

Strings and Text Files

To be competent, the user/individual on the job must be able to:

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- PC14.** Access and manipulate strings using methods like slicing, testing, searching, and basic string operations.
- PC15.** Implement string methods to process and manipulate text efficiently.
- PC16.** Perform file operations, including reading from and writing to text files, and apply basic data encryption techniques.
- PC17.** Work with lists, including slicing, finding items, and processing two-dimensional lists.
- PC18.** Use dictionaries and sets for data organization and demonstrate the serialization of objects.
- PC19.** Explain the concept of recursion and solve problems using recursive algorithms.

Design with Classes

To be competent, the user/individual on the job must be able to:

- PC20.** Define and create classes and objects, applying object-oriented principles such as inheritance and polymorphism.
- PC21.** Implement methods within classes and demonstrate the interaction between classes and functions.
- PC22.** Utilize techniques for designing classes, ensuring proper encapsulation and modularity in code design.
- PC23.** Manage instances of classes, applying object-oriented techniques to solve complex problems.

Graphical User Interfaces (GUI)

To be competent, the user/individual on the job must be able to:

- PC24.** Differentiate between terminal-based and GUI-based programs, and code simple GUI-based applications using Python.
- PC25.** Utilize the Tkinter interface to create graphical user interfaces, including widgets like labels, buttons, and entry fields.
- PC26.** Implement simple graphics using Turtle Graphics, managing two-dimensional shapes and colours.
- PC27.** Perform basic image processing tasks, utilizing the RGB color system to manipulate images.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the basic syntax, variables, and data types in Python programming.
- KU2.** Recognize the use of Python for data manipulation, automation, and business analytics.
- KU3.** Explain key programming concepts such as loops, functions, and conditional statements.
- KU4.** Identify common Python libraries used for data analysis, including Pandas, NumPy, and Matplotlib.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate programming concepts and logic effectively in business contexts.
- GS2.** Apply problem-solving skills to write and debug Python scripts for data processing.

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- GS3.** Use critical thinking to select appropriate Python functions for specific tasks.
- GS4.** Demonstrate proficiency in Python coding and the use of Jupyter Notebook or other Python development environments.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Python</i>	10	2	-	-
PC1. Install Python on various operating systems and set up the programming environment.	-	-	-	-
PC2. Explain how a program works and describe the program development cycle.	-	-	-	-
PC3. Demonstrate basic input, processing, and output operations using Python, including the use of the print function and reading input from the keyboard.	-	-	-	-
PC4. Implement basic calculations and expressions, utilizing operators and type conversions effectively.	-	-	-	-
PC5. Apply decision structures using if, if-else, and nested decision-making in Python.	-	-	-	-
PC6. Use logical operators and Boolean variables to perform conditional checks.	-	-	-	-
PC7. Implement repetition structures using while and for loops, and demonstrate the use of loops for running totals and input validation.	-	-	-	-
PC8. Work with strings, numeric data types, and character sets, utilizing functions and modules to manipulate and process data.	-	-	-	-
<i>Control Statements</i>	20	2	-	-
PC9. Implement definite and conditional iteration in Python programs, formatting text for output appropriately.	-	-	-	-
PC10. Handle file input and output operations, including processing records and managing exceptions during file handling.	-	-	-	-
PC11. Define and call functions, including value-returning functions, and demonstrate the use of local and global variables.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Utilize Python's math module and random number generation within custom functions.	-	-	-	-
PC13. Organize code by storing functions in modules, ensuring modular design and reusability.	-	-	-	-
<i>Strings and Text Files</i>	20	2	-	-
PC14. Access and manipulate strings using methods like slicing, testing, searching, and basic string operations.	-	-	-	-
PC15. Implement string methods to process and manipulate text efficiently.	-	-	-	-
PC16. Perform file operations, including reading from and writing to text files, and apply basic data encryption techniques.	-	-	-	-
PC17. Work with lists, including slicing, finding items, and processing two-dimensional lists.	-	-	-	-
PC18. Use dictionaries and sets for data organization and demonstrate the serialization of objects.	-	-	-	-
PC19. Explain the concept of recursion and solve problems using recursive algorithms.	-	-	-	-
<i>Design with Classes</i>	20	2	-	-
PC20. Define and create classes and objects, applying object-oriented principles such as inheritance and polymorphism.	-	-	-	-
PC21. Implement methods within classes and demonstrate the interaction between classes and functions.	-	-	-	-
PC22. Utilize techniques for designing classes, ensuring proper encapsulation and modularity in code design.	-	-	-	-
PC23. Manage instances of classes, applying object-oriented techniques to solve complex problems.	-	-	-	-
<i>Graphical User Interfaces (GUI)</i>	20	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. Differentiate between terminal-based and GUI-based programs, and code simple GUI-based applications using Python.	-	-	-	-
PC25. Utilize the TKinter interface to create graphical user interfaces, including widgets like labels, buttons, and entry fields.	-	-	-	-
PC26. Implement simple graphics using Turtle Graphics, managing two-dimensional shapes and colours.	-	-	-	-
PC27. Perform basic image processing tasks, utilizing the RGB color system to manipulate images.	-	-	-	-
NOS Total	90	10	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MEP/N1520
NOS Name	Basics of Python
Sector	Management
Sub-Sector	
Occupation	Operations
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024

Qualification Pack

MEP/N1521: Data Warehousing and OLAP

Description

This NOS focuses on the concepts of data warehousing and Online Analytical Processing (OLAP) to support efficient data storage, retrieval, and business intelligence for decision-making.

Scope

The scope covers the following :

- Introduction to Data Warehousing
- Planning a Data Warehouse
- Data Warehouse Schema and Modelling
- ETL Process and Maintenance of Data Warehouse
- OLAP in Data Warehouse

Elements and Performance Criteria

Introduction to Data Warehousing

To be competent, the user/individual on the job must be able to:

- PC1.** Define data warehousing and explain its benefits in managing and analyzing large datasets.
- PC2.** Identify and describe the key features of a data warehouse, including data granularity and the information flow mechanism.
- PC3.** Explain the importance of metadata in a data warehouse and describe its role in data management and retrieval.
- PC4.** Outline the architecture of a data warehouse and explain the concept of data marts, including the process of building data marts and addressing related issues.

Planning a Data Warehouse

To be competent, the user/individual on the job must be able to:

- PC5.** Determine the end-user requirements for a data warehouse using various requirement-gathering methods.
- PC6.** Conduct requirement analysis and dimensional analysis to structure the data warehouse effectively.
- PC7.** Develop Information Package Diagrams (IPD) to visualize data flow and relationships within the warehouse.
- PC8.** Create a comprehensive data warehouse project plan, including an economic feasibility analysis to assess the project's viability.

Data Warehouse Schema and Modelling

To be competent, the user/individual on the job must be able to:

- PC9.** Apply dimensional modelling techniques to design data warehouse schemas, including star schema and snowflake schema.
- PC10.** Create and manage aggregate tables and fact constellation tables to optimize query performance.

Qualification Pack

PC11. Differentiate between dimensional tables and fact tables and explain the role of keys in maintaining data integrity within the warehouse.

ETL Process and Maintenance of Data Warehouse

To be competent, the user/individual on the job must be able to:

PC12. Perform data extraction, transformation, and loading (ETL) processes to populate the data warehouse with accurate and relevant data.

PC13. Ensure data quality through rigorous design reviews, testing, and monitoring of the data warehouse.

PC14. Implement strategies for ongoing maintenance of the data warehouse to ensure optimal performance and data accuracy.

OLAP in Data Warehouse

To be competent, the user/individual on the job must be able to:

PC15. Explain the concept of OLAP (Online Analytical Processing) and its role in enabling multidimensional analysis within a data warehouse.

PC16. Apply OLAP functions and models (MOLAP, ROLAP, HOLAP, DOLAP) to perform complex data analysis and support decision-making processes.

PC17. Design OLAP solutions, considering key factors such as data structure, query performance, and user requirements.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the fundamentals of data warehousing, including ETL (Extract, Transform, Load) processes and data integration.

KU2. Recognize the role of OLAP in multidimensional data analysis and reporting.

KU3. Explain data modeling techniques, including star and snowflake schemas, for structuring a data warehouse.

KU4. Identify best practices for optimizing data storage, retrieval, and query performance in OLAP systems.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Communicate data warehousing concepts and analytical insights effectively to stakeholders.

GS2. Apply analytical skills to design and manage OLAP cubes for business intelligence.

GS3. Use problem-solving techniques to optimize data processing and warehouse performance.

GS4. Demonstrate proficiency in data warehousing tools and OLAP solutions such as SQL, Power BI, and cloud-based platforms.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Data Warehousing</i>	10	4	-	-
PC1. Define data warehousing and explain its benefits in managing and analyzing large datasets.	-	-	-	-
PC2. Identify and describe the key features of a data warehouse, including data granularity and the information flow mechanism.	-	-	-	-
PC3. Explain the importance of metadata in a data warehouse and describe its role in data management and retrieval.	-	-	-	-
PC4. Outline the architecture of a data warehouse and explain the concept of data marts, including the process of building data marts and addressing related issues.	-	-	-	-
<i>Planning a Data Warehouse</i>	10	4	-	-
PC5. Determine the end-user requirements for a data warehouse using various requirement-gathering methods.	-	-	-	-
PC6. Conduct requirement analysis and dimensional analysis to structure the data warehouse effectively.	-	-	-	-
PC7. Develop Information Package Diagrams (IPD) to visualize data flow and relationships within the warehouse.	-	-	-	-
PC8. Create a comprehensive data warehouse project plan, including an economic feasibility analysis to assess the project's viability.	-	-	-	-
<i>Data Warehouse Schema and Modelling</i>	20	4	-	-
PC9. Apply dimensional modelling techniques to design data warehouse schemas, including star schema and snowflake schema.	-	-	-	-
PC10. Create and manage aggregate tables and fact constellation tables to optimize query performance.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Differentiate between dimensional tables and fact tables and explain the role of keys in maintaining data integrity within the warehouse.	-	-	-	-
<i>ETL Process and Maintenance of Data Warehouse</i>	20	4	-	-
PC12. Perform data extraction, transformation, and loading (ETL) processes to populate the data warehouse with accurate and relevant data.	-	-	-	-
PC13. Ensure data quality through rigorous design reviews, testing, and monitoring of the data warehouse.	-	-	-	-
PC14. Implement strategies for ongoing maintenance of the data warehouse to ensure optimal performance and data accuracy.	-	-	-	-
<i>OLAP in Data Warehouse</i>	20	4	-	-
PC15. Explain the concept of OLAP (Online Analytical Processing) and its role in enabling multidimensional analysis within a data warehouse.	-	-	-	-
PC16. Apply OLAP functions and models (MOLAP, ROLAP, HOLAP, DOLAP) to perform complex data analysis and support decision-making processes.	-	-	-	-
PC17. Design OLAP solutions, considering key factors such as data structure, query performance, and user requirements.	-	-	-	-
NOS Total	80	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MEP/N1521
NOS Name	Data Warehousing and OLAP
Sector	Management
Sub-Sector	
Occupation	Operations
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQF Clearance Date	17/12/2024

Qualification Pack

MEP/N1522: Big Data Analytics

Description

This NOS focuses on processing, analyzing, and interpreting large-scale datasets to extract meaningful insights, optimize business strategies, and support data-driven decision-making.

Scope

The scope covers the following :

- Introduction to Big Data
- Hadoop MapReduce
- Introduction to Apache PIG
- Apache Hive
- Apache HBase

Elements and Performance Criteria

Introduction to Big Data

To be competent, the user/individual on the job must be able to:

- PC1.** Define the concept of big data and explain its key features, including volume, velocity, variety, and veracity.
- PC2.** Identify and describe the challenges associated with managing and analysing big data in modern enterprises.
- PC3.** Explain the features of Hadoop and its role in big data processing.
- PC4.** Describe the components and architecture of the Hadoop ecosystem, including Hadoop Distributed File System (HDFS) and Hadoop clusters.
- PC5.** Demonstrate understanding of Hadoop architecture, including HDFS for distributed storage and data management within a Hadoop cluster.

Hadoop MapReduce

To be competent, the user/individual on the job must be able to:

- PC6.** Explain the concept of MapReduce and its importance in processing large datasets within the Hadoop framework.
- PC7.** Describe the YARN components and architecture, explaining how YARN manages resources in a Hadoop cluster.
- PC8.** Outline the execution flow of a YARN MapReduce application, detailing the stages of the YARN workflow.
- PC9.** Develop and analyze a basic MapReduce program, including the use of distributed cache, MRUnit for testing, Reduce Join, and custom input formats.

Introduction to Apache PIG

To be competent, the user/individual on the job must be able to:

- PC10.** Explain the components of Apache PIG and describe its execution environment within the Hadoop ecosystem.

Qualification Pack

- PC11.** Identify and work with various PIG data types and data models to process and analyze large datasets.
- PC12.** Develop PIG scripts for data processing, utilizing PIG's data flow language to transform and analyze data efficiently.

Apache Hive

To be competent, the user/individual on the job must be able to:

- PC13.** Describe the architecture and components of Apache HIVE, explaining its role in data warehousing within Hadoop.
- PC14.** Identify HIVE data types and models and demonstrate how to use HIVE partitioning and bucketing for optimized data management.
- PC15.** Create and manage HIVE tables, using HIVE QL to join tables and perform dynamic partitioning.

Apache HBase

To be competent, the user/individual on the job must be able to:

- PC16.** Introduce the architecture and components of Apache HBase, including its run modes and configuration options.
- PC17.** Explain the data models used in HBase and demonstrate HBase data loading techniques.
- PC18.** Provide an overview of Apache Spark, describing its integration with Hadoop and its capabilities for fast, in-memory data processing.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the fundamentals of big data, including volume, velocity, variety, and veracity (4Vs).
- KU2.** Recognize the role of big data technologies such as Hadoop, Spark, and cloud-based analytics platforms.
- KU3.** Explain data processing techniques, including data ingestion, storage, and real-time analytics.
- KU4.** Identify best practices for big data security, privacy, and compliance.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate big data insights effectively using visualizations and reports.
- GS2.** Apply analytical skills to process and interpret large datasets for business intelligence.
- GS3.** Use problem-solving techniques to optimize data workflows and enhance performance.
- GS4.** Demonstrate proficiency in big data tools such as Python, R, SQL, and cloud-based analytics platforms.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Big Data</i>	10	4	-	-
PC1. Define the concept of big data and explain its key features, including volume, velocity, variety, and veracity.	-	-	-	-
PC2. Identify and describe the challenges associated with managing and analysing big data in modern enterprises.	-	-	-	-
PC3. Explain the features of Hadoop and its role in big data processing.	-	-	-	-
PC4. Describe the components and architecture of the Hadoop ecosystem, including Hadoop Distributed File System (HDFS) and Hadoop clusters.	-	-	-	-
PC5. Demonstrate understanding of Hadoop architecture, including HDFS for distributed storage and data management within a Hadoop cluster.	-	-	-	-
<i>Hadoop MapReduce</i>	20	4	-	-
PC6. Explain the concept of MapReduce and its importance in processing large datasets within the Hadoop framework.	-	-	-	-
PC7. Describe the YARN components and architecture, explaining how YARN manages resources in a Hadoop cluster.	-	-	-	-
PC8. Outline the execution flow of a YARN MapReduce application, detailing the stages of the YARN workflow.	-	-	-	-
PC9. Develop and analyze a basic MapReduce program, including the use of distributed cache, MRUnit for testing, Reduce Join, and custom input formats.	-	-	-	-
<i>Introduction to Apache PIG</i>	20	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Explain the components of Apache PIG and describe its execution environment within the Hadoop ecosystem.	-	-	-	-
PC11. Identify and work with various PIG data types and data models to process and analyze large datasets.	-	-	-	-
PC12. Develop PIG scripts for data processing, utilizing PIG's data flow language to transform and analyze data efficiently.	-	-	-	-
<i>Apache Hive</i>	10	4	-	-
PC13. Describe the architecture and components of Apache HIVE, explaining its role in data warehousing within Hadoop.	-	-	-	-
PC14. Identify HIVE data types and models and demonstrate how to use HIVE partitioning and bucketing for optimized data management.	-	-	-	-
PC15. Create and manage HIVE tables, using HIVE QL to join tables and perform dynamic partitioning.	-	-	-	-
<i>Apache HBase</i>	20	4	-	-
PC16. Introduce the architecture and components of Apache HBase, including its run modes and configuration options.	-	-	-	-
PC17. Explain the data models used in HBase and demonstrate HBase data loading techniques.	-	-	-	-
PC18. Provide an overview of Apache Spark, describing its integration with Hadoop and its capabilities for fast, in-memory data processing.	-	-	-	-
NOS Total	80	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MEP/N1522
NOS Name	Big Data Analytics
Sector	Management
Sub-Sector	
Occupation	Operations
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024

Qualification Pack

MEP/N1523: OJT Mandatory - Business Analytics

Description

This NOS focuses on providing hands-on experience in business analytics, enabling learners to apply data analysis techniques, statistical tools, and business intelligence concepts in real-world scenarios.

Scope

The scope covers the following :

- OJT Mandatory - Business Analytics

Elements and Performance Criteria

OJT Mandatory - Business Analytics

To be competent, the user/individual on the job must be able to:

- PC1.** Assess data quality of datasets and apply data-cleaning techniques to handle missing or incomplete data.
- PC2.** Perform data mining techniques (e.g., association rules, clustering) using business datasets.
- PC3.** Conduct regression analysis using relevant software (e.g., Excel, R) to establish relationships between variables and predict outcomes.
- PC4.** Solve probability problems using addition and multiplication theorems relevant to the business scenario of the organisation
- PC5.** Perform a workshop on setting business goals and selecting KPIs for a social media campaign.
- PC6.** Use Google Analytics to track and analyse web pages of the organisation and present the results.
- PC7.** Calculate and interpret consumer surplus across different market scenarios.
- PC8.** Demonstrate the use of Big Data for a project
- PC9.** Deliver a presentation on a selected business topic to the team
- PC10.** Draft a social blog on a business topic, considering audience and platform appropriateness.
- PC11.** Research and prepare a note on a relevant business topic for the organisation
- PC12.** Use the Power BI Report Designer to organize report visuals, pages, and multiple data sources.
- PC13.** Design Power BI visualizations, utilizing the canvas, visualizations pane, and fields pane.
- PC14.** Create charts in Power BI (e.g., bar, line, scatter) and apply customization options like axis settings and tooltips.
- PC15.** Support the management in a corporate decision-making in respect to the current macro-economic plans in India
- PC16.** Demonstrate the use of Excel for creating graphs, tables, and spreadsheets to interpret HR data.
- PC17.** Apply auto-regression techniques to time series data in R to enhance forecasting accuracy.

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- PC18.** Create a spreadsheet using various formulas and functions to analyse financial or statistical data.
- PC19.** Send a report to different teams in the organisation using the feature mail merge
- PC20.** Demonstrate appropriate professional etiquettes such as respectful conduct, proper attire, and professional communication.
- PC21.** Participate in group discussions on various topics, practicing respectful interaction, clear articulation, and constructive dialogue.
- PC22.** Utilize virtual collaboration tools to work effectively
- PC23.** Create arrays and perform element-wise arithmetic operations.
- PC24.** Create data frames, manipulate components, aggregate and sort data, and demonstrate merging, reshaping, and sub-setting data frames.
- PC25.** Generate reports in MS Access, including summary reports and crosstab reports, to present data effectively.
- PC26.** Perform ER analysis, defining attributes and domains in SQL, and apply primary keys, foreign keys, and unique constraints to ensure data integrity.
- PC27.** Connect to Excel, text files, and Microsoft Analysis Services within Tableau, importing data for report generation.
- PC28.** Create hierarchies and parameters in Tableau to enable drill-down analysis and dynamic control of visualizations.
- PC29.** Apply forecasting techniques to predict supply chain requirements using real data.
- PC30.** Develop and implement models for Supplier Relationship Management (SRM) using data provided by the organisation
- PC31.** Use provided data to develop a marketing strategy targeting a specific audience to maximize marketing ROI.
- PC32.** Calculate demand curves and maximum WTP using data provided by the organisation and develop pricing strategies based on value pricing principles.
- PC33.** Use advanced functions (e.g., IF, VLOOKUP, INDEX-MATCH) for specific data analysis tasks.
- PC34.** Apply conditional formatting rules to emphasize key data points, using colour scales and data bars.
- PC35.** Practice conversations in various scenarios, using practical tips to improve conversational effectiveness, tone, and body language.
- PC36.** Develop small programs to process data using loops and conditional statements in Python.
- PC37.** Create classes with attributes and methods and demonstrate inheritance and polymorphism.
- PC38.** Build GUI-based applications using Tkinter to build basic interactive design.
- PC39.** Design a basic blueprint of a data warehouse, labelling the components and explaining data flow.
- PC40.** Develop a simplified Information Package Diagram (IPD) to represent data relationships for a warehouse.
- PC41.** Design basic OLAP queries, using models like MOLAP and ROLAP, to analyse class data.
- PC42.** Evaluate query performance using different OLAP models and optimize based on data structure.
- PC43.** Outline the execution flow of a YARN MapReduce application, detailing each stage.
- PC44.** Process large datasets using different PIG data types and data models

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- PC45.** Install and configure Hadoop to set up HDFS and a functioning Hadoop cluster.
- PC46.** Execute a MapReduce program on a sample dataset to demonstrate its functionality in large-scale data processing.
- PC47.** Design an Apache Spark application to process data in-memory and evaluate its performance impact.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the practical applications of business analytics in various industries, including finance, marketing, and operations.
- KU2.** Recognize key data analysis techniques, visualization tools, and predictive modeling methods used in business analytics.
- KU3.** Explain the importance of data-driven decision-making and its impact on business performance.
- KU4.** Identify best practices for handling real-world data, ensuring accuracy, compliance, and ethical considerations.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate analytical insights effectively through reports, dashboards, and presentations.
- GS2.** Apply problem-solving skills to analyze business challenges using data-driven approaches.
- GS3.** Use critical thinking to interpret data trends and recommend actionable strategies.
- GS4.** Demonstrate proficiency in business analytics tools such as Excel, Power BI, SQL, and Python.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>OJT Mandatory - Business Analytics</i>	-	100	100	100
PC1. Assess data quality of datasets and apply data-cleaning techniques to handle missing or incomplete data.	-	-	-	-
PC2. Perform data mining techniques (e.g., association rules, clustering) using business datasets.	-	-	-	-
PC3. Conduct regression analysis using relevant software (e.g., Excel, R) to establish relationships between variables and predict outcomes.	-	-	-	-
PC4. Solve probability problems using addition and multiplication theorems relevant to the business scenario of the organisation	-	-	-	-
PC5. Perform a workshop on setting business goals and selecting KPIs for a social media campaign.	-	-	-	-
PC6. Use Google Analytics to track and analyse web pages of the organisation and present the results.	-	-	-	-
PC7. Calculate and interpret consumer surplus across different market scenarios.	-	-	-	-
PC8. Demonstrate the use of Big Data for a project	-	-	-	-
PC9. Deliver a presentation on a selected business topic to the team	-	-	-	-
PC10. Draft a social blog on a business topic, considering audience and platform appropriateness.	-	-	-	-
PC11. Research and prepare a note on a relevant business topic for the organisation	-	-	-	-
PC12. Use the Power BI Report Designer to organize report visuals, pages, and multiple data sources.	-	-	-	-
PC13. Design Power BI visualizations, utilizing the canvas, visualizations pane, and fields pane.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Create charts in Power BI (e.g., bar, line, scatter) and apply customization options like axis settings and tooltips.	-	-	-	-
PC15. Support the management in a corporate decision-making in respect to the current macro-economic plans in India	-	-	-	-
PC16. Demonstrate the use of Excel for creating graphs, tables, and spreadsheets to interpret HR data.	-	-	-	-
PC17. Apply auto-regression techniques to time series data in R to enhance forecasting accuracy.	-	-	-	-
PC18. Create a spreadsheet using various formulas and functions to analyse financial or statistical data.	-	-	-	-
PC19. Send a report to different teams in the organisation using the feature mail merge	-	-	-	-
PC20. Demonstrate appropriate professional etiquettes such as respectful conduct, proper attire, and professional communication.	-	-	-	-
PC21. Participate in group discussions on various topics, practicing respectful interaction, clear articulation, and constructive dialogue.	-	-	-	-
PC22. Utilize virtual collaboration tools to work effectively	-	-	-	-
PC23. Create arrays and perform element-wise arithmetic operations.	-	-	-	-
PC24. Create data frames, manipulate components, aggregate and sort data, and demonstrate merging, reshaping, and sub-setting data frames.	-	-	-	-
PC25. Generate reports in MS Access, including summary reports and crosstab reports, to present data effectively.	-	-	-	-
PC26. Perform ER analysis, defining attributes and domains in SQL, and apply primary keys, foreign keys, and unique constraints to ensure data integrity.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. Connect to Excel, text files, and Microsoft Analysis Services within Tableau, importing data for report generation.	-	-	-	-
PC28. Create hierarchies and parameters in Tableau to enable drill-down analysis and dynamic control of visualizations.	-	-	-	-
PC29. Apply forecasting techniques to predict supply chain requirements using real data.	-	-	-	-
PC30. Develop and implement models for Supplier Relationship Management (SRM) using data provided by the organisation	-	-	-	-
PC31. Use provided data to develop a marketing strategy targeting a specific audience to maximize marketing ROI.	-	-	-	-
PC32. Calculate demand curves and maximum WTP using data provided by the organisation and develop pricing strategies based on value pricing principles.	-	-	-	-
PC33. Use advanced functions (e.g., IF, VLOOKUP, INDEX-MATCH) for specific data analysis tasks.	-	-	-	-
PC34. Apply conditional formatting rules to emphasize key data points, using colour scales and data bars.	-	-	-	-
PC35. Practice conversations in various scenarios, using practical tips to improve conversational effectiveness, tone, and body language.	-	-	-	-
PC36. Develop small programs to process data using loops and conditional statements in Python.	-	-	-	-
PC37. Create classes with attributes and methods and demonstrate inheritance and polymorphism.	-	-	-	-
PC38. Build GUI-based applications using TKinter to build basic interactive design.	-	-	-	-
PC39. Design a basic blueprint of a data warehouse, labelling the components and explaining data flow.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC40. Develop a simplified Information Package Diagram (IPD) to represent data relationships for a warehouse.	-	-	-	-
PC41. Design basic OLAP queries, using models like MOLAP and ROLAP, to analyse class data.	-	-	-	-
PC42. Evaluate query performance using different OLAP models and optimize based on data structure.	-	-	-	-
PC43. Outline the execution flow of a YARN MapReduce application, detailing each stage.	-	-	-	-
PC44. Process large datasets using different PIG data types and data models	-	-	-	-
PC45. Install and configure Hadoop to set up HDFS and a functioning Hadoop cluster.	-	-	-	-
PC46. Execute a MapReduce program on a sample dataset to demonstrate its functionality in large-scale data processing.	-	-	-	-
PC47. Design an Apache Spark application to process data in-memory and evaluate its performance impact.	-	-	-	-
NOS Total	-	100	100	100

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MEP/N1523
NOS Name	OJT Mandatory - Business Analytics
Sector	Management
Sub-Sector	
Occupation	Operations
NSQF Level	5
Credits	10
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024

Qualification Pack

MEP/N1518: Logistics and Supply Chain Analytics

Description

This NOS focuses on leveraging data analytics to optimize logistics and supply chain operations, enhance efficiency, reduce costs, and improve decision-making.

Scope

The scope covers the following :

- Introduction to Supply Chain Analytics
- Overview of Supply Chain Models and Modeling Systems
- Logistics Models
- Scheduling
- Collaboration

Elements and Performance Criteria

Introduction to Supply Chain Analytics

To be competent, the user/individual on the job must be able to:

- PC1.** Define supply chain analytics and explain its relevance, significance, and scope in modern supply chain management.
- PC2.** Identify and describe the various models and techniques used in supply chain analytics, highlighting their applications in optimizing supply chain processes.

Overview of Supply Chain Models and Modeling Systems

To be competent, the user/individual on the job must be able to:

- PC3.** Explain the concept of linear programming and its application in supply chain management.
- PC4.** Analyze the advantages and disadvantages of using linear programming models in optimizing supply chain operations.

Logistics Models

To be competent, the user/individual on the job must be able to:

- PC5.** Apply forecasting techniques and demand management strategies to predict and manage supply chain requirements.
- PC6.** Implement capacity planning techniques to ensure optimal resource allocation in supply chain operations.
- PC7.** Utilize inventory management techniques, including MRP I & II, to maintain optimal inventory levels.
- PC8.** Apply procurement and sourcing techniques to streamline the acquisition of goods and services within the supply chain.

Scheduling

To be competent, the user/individual on the job must be able to:

- PC9.** Describe various supply chain operations and apply scheduling techniques to optimize the flow of goods and services.

Qualification Pack

- PC10.** Implement models for Supplier Relationship Management (SRM) to enhance collaboration and efficiency in the supply chain.
- PC11.** Apply queuing theory to supply chain processes, analyzing its advantages and disadvantages in managing wait times and optimizing service levels.

Collaboration

To be competent, the user/individual on the job must be able to:

- PC12.** Implement Collaborative Planning, Forecasting, and Replenishment (CPFR) models to improve coordination across the supply chain.
- PC13.** Develop strategies for effective customer service by aligning supply chain operations with customer expectations.
- PC14.** Apply logistics models, including transportation and assignment models, to optimize the movement of goods and reduce operational costs.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the key concepts of logistics and supply chain management, including procurement, inventory, and distribution.
- KU2.** Recognize the role of data analytics in forecasting demand, optimizing routes, and improving supply chain performance.
- KU3.** Explain key performance indicators (KPIs) used in logistics and supply chain analytics.
- KU4.** Identify best practices for supply chain risk management and cost optimization using analytical tools.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate supply chain insights effectively using reports and dashboards.
- GS2.** Apply analytical skills to interpret supply chain data and enhance operational efficiency.
- GS3.** Use problem-solving techniques to address supply chain bottlenecks and optimize logistics processes.
- GS4.** Demonstrate proficiency in supply chain analytics tools such as Excel, Power BI, and specialized logistics software.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Supply Chain Analytics</i>	10	-	4	-
PC1. Define supply chain analytics and explain its relevance, significance, and scope in modern supply chain management.	-	-	-	-
PC2. Identify and describe the various models and techniques used in supply chain analytics, highlighting their applications in optimizing supply chain processes.	-	-	-	-
<i>Overview of Supply Chain Models and Modeling Systems</i>	20	-	4	-
PC3. Explain the concept of linear programming and its application in supply chain management.	-	-	-	-
PC4. Analyze the advantages and disadvantages of using linear programming models in optimizing supply chain operations.	-	-	-	-
<i>Logistics Models</i>	20	-	4	-
PC5. Apply forecasting techniques and demand management strategies to predict and manage supply chain requirements.	-	-	-	-
PC6. Implement capacity planning techniques to ensure optimal resource allocation in supply chain operations.	-	-	-	-
PC7. Utilize inventory management techniques, including MRP I & II, to maintain optimal inventory levels.	-	-	-	-
PC8. Apply procurement and sourcing techniques to streamline the acquisition of goods and services within the supply chain.	-	-	-	-
<i>Scheduling</i>	20	-	4	-
PC9. Describe various supply chain operations and apply scheduling techniques to optimize the flow of goods and services.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Implement models for Supplier Relationship Management (SRM) to enhance collaboration and efficiency in the supply chain.	-	-	-	-
PC11. Apply queuing theory to supply chain processes, analyzing its advantages and disadvantages in managing wait times and optimizing service levels.	-	-	-	-
<i>Collaboration</i>	10	-	4	-
PC12. Implement Collaborative Planning, Forecasting, and Replenishment (CPFR) models to improve coordination across the supply chain.	-	-	-	-
PC13. Develop strategies for effective customer service by aligning supply chain operations with customer expectations.	-	-	-	-
PC14. Apply logistics models, including transportation and assignment models, to optimize the movement of goods and reduce operational costs.	-	-	-	-
NOS Total	80	-	20	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MEP/N1518
NOS Name	Logistics and Supply Chain Analytics
Sector	Management
Sub-Sector	
Occupation	Operations
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024

Qualification Pack

MEP/N1519: Marketing and Retail Analytics

Description

This NOS focuses on applying data analytics to optimize marketing strategies, enhance customer insights, and improve retail performance through data-driven decision-making.

Scope

The scope covers the following :

- Marketing Process
- Overview of Retailing
- Pricing tactics
- Measuring Price and Promotion Response
- Retailers Location Decision

Elements and Performance Criteria

Marketing Process

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the importance of marketing analytics and its role in enhancing marketing strategies and decision-making.
- PC2.** Describe the fundamental steps in the marketing process and how data can be utilized to optimize each step.
- PC3.** Develop a marketing strategy using data-driven insights to target the right audience and maximize marketing ROI.
- PC4.** Implement analytics to continuously improve the marketing process, making data-informed adjustments to strategies and campaigns.

Overview of Retailing

To be competent, the user/individual on the job must be able to:

- PC5.** Analyse the modern retailing marketplace, focusing on technological advancements and their impact on retail strategies.
- PC6.** Utilize data visualization techniques to represent retail data effectively, aiding in the interpretation of trends and patterns.
- PC7.** Apply descriptive analysis methods to retail data, providing insights into customer behaviour, sales trends, and inventory management.

Pricing tactics

To be competent, the user/individual on the job must be able to:

- PC8.** Explain the principles of value pricing, including the net-benefit principle, and apply these concepts to real-world pricing strategies.
- PC9.** Analyse demand curves to determine maximum Willingness to Pay (WTP) and develop pricing strategies that maximize customer value.
- PC10.** Identify competitive advantages in pricing strategies and apply linear models, such as simple regressions, to analyse pricing data.

Qualification Pack

Measuring Price and Promotion Response

To be competent, the user/individual on the job must be able to:

- PC11.** Apply price theory concepts, including elasticities, to analyse the impact of pricing decisions on profit and revenue maximization.
- PC12.** Use log-log models and multivariate regressions to measure the response of sales to price changes and promotional activities.
- PC13.** Develop effective promotion strategies using game theory principles, analysing simultaneous and sequential-move games to anticipate competitor behaviour.

Retailers' Location Decision

To be competent, the user/individual on the job must be able to:

- PC14.** Apply machine learning techniques, such as decision tree or random forest, to decide on a retail location.
- PC15.** Utilize the Huff model and regression models to determine optimal retail locations, considering factors like economies of scale and scope.
- PC16.** Analyse repeated games and tactical collusion in retail location decisions to maximize competitive advantage.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the key concepts of marketing and retail analytics, including customer segmentation, pricing strategies, and sales forecasting.
- KU2.** Recognize the role of data analytics in consumer behavior analysis and targeted marketing campaigns.
- KU3.** Explain the use of key performance indicators (KPIs) such as customer lifetime value (CLV), conversion rates, and sales trends.
- KU4.** Identify best practices for using analytics tools to optimize marketing ROI and retail operations.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate marketing and retail insights effectively through reports and data visualizations.
- GS2.** Apply analytical skills to interpret consumer data and develop data-driven marketing strategies.
- GS3.** Use problem-solving techniques to optimize pricing, promotions, and customer engagement strategies.
- GS4.** Demonstrate proficiency in analytics tools such as Google Analytics, Power BI, and CRM software.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Marketing Process</i>	10	-	4	-
PC1. Explain the importance of marketing analytics and its role in enhancing marketing strategies and decision-making.	-	-	-	-
PC2. Describe the fundamental steps in the marketing process and how data can be utilized to optimize each step.	-	-	-	-
PC3. Develop a marketing strategy using data-driven insights to target the right audience and maximize marketing ROI.	-	-	-	-
PC4. Implement analytics to continuously improve the marketing process, making data-informed adjustments to strategies and campaigns.	-	-	-	-
<i>Overview of Retailing</i>	20	-	4	-
PC5. Analyse the modern retailing marketplace, focusing on technological advancements and their impact on retail strategies.	-	-	-	-
PC6. Utilize data visualization techniques to represent retail data effectively, aiding in the interpretation of trends and patterns.	-	-	-	-
PC7. Apply descriptive analysis methods to retail data, providing insights into customer behaviour, sales trends, and inventory management.	-	-	-	-
<i>Pricing tactics</i>	20	-	4	-
PC8. Explain the principles of value pricing, including the net-benefit principle, and apply these concepts to real-world pricing strategies.	-	-	-	-
PC9. Analyse demand curves to determine maximum Willingness to Pay (WTP) and develop pricing strategies that maximize customer value.	-	-	-	-
PC10. Identify competitive advantages in pricing strategies and apply linear models, such as simple regressions, to analyse pricing data.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Measuring Price and Promotion Response</i>	20	-	4	-
PC11. Apply price theory concepts, including elasticities, to analyse the impact of pricing decisions on profit and revenue maximization.	-	-	-	-
PC12. Use log-log models and multivariate regressions to measure the response of sales to price changes and promotional activities.	-	-	-	-
PC13. Develop effective promotion strategies using game theory principles, analysing simultaneous and sequential-move games to anticipate competitor behaviour.	-	-	-	-
<i>Retailers' Location Decision</i>	10	-	4	-
PC14. Apply machine learning techniques, such as decision tree or random forest, to decide on a retail location.	-	-	-	-
PC15. Utilize the Huff model and regression models to determine optimal retail locations, considering factors like economies of scale and scope.	-	-	-	-
PC16. Analyse repeated games and tactical collusion in retail location decisions to maximize competitive advantage.	-	-	-	-
NOS Total	80	-	20	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MEP/N1519
NOS Name	Marketing and Retail Analytics
Sector	Management
Sub-Sector	
Occupation	Operations
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score recommended pass % aggregate for the QP.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MEP/N1515.Basics of R Programming	90	10	-	-	100	10
MEP/N1516.Database Management System	90	10	-	-	100	10
MEP/N1517.Data Visualization Tools - Tableau	80	20	-	-	100	10
MEP/N0259.Excel for Business Applications	90	10	-	-	100	10
MEP/N0256.Personality Development	80	20	-	-	100	5
MEP/N0422.Entrepreneurship Development & Innovation (EDI)	100	-	-	-	100	5
MEP/N1520.Basics of Python	90	10	-	-	100	10
MEP/N1521.Data Warehousing and OLAP	80	20	-	-	100	10
MEP/N1522.Big Data Analytics	80	20	-	-	100	10
MEP/N1523.OJT Mandatory - Business Analytics	-	100	100	100	300	10
Total	780	220	100	100	1200	90

Elective: 1 Logistics and Supply Chain Analytics

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MEP/N1518.Logistics and Supply Chain Analytics	80	-	20	-	100	10
Total	80	0	20	-	100	10

Elective: 2 Marketing & Retail Analytics

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MEP/N1519.Marketing and Retail Analytics	80	-	20	-	100	10
Total	80	0	20	-	100	10

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.